

Peripherals and Interfaces

2020-2021

4[B] Interface two chips of 4k RAM and two chips of 2k ROM with processor. Starting address of RAM is A000H. Draw a neat logic diagram of your implementation. (5)

4[D] Define the three types of VO. Identify each as either microprocessor-initiated or device-initiated. (2)

5[C] Summarize the basic difference between maskable and non-maskable interrupts. Describe how power failure interrupt is normally handled. (3)

6[C] List the methods used in main memory array design. What are the advantages and disadvantages of each? (3)

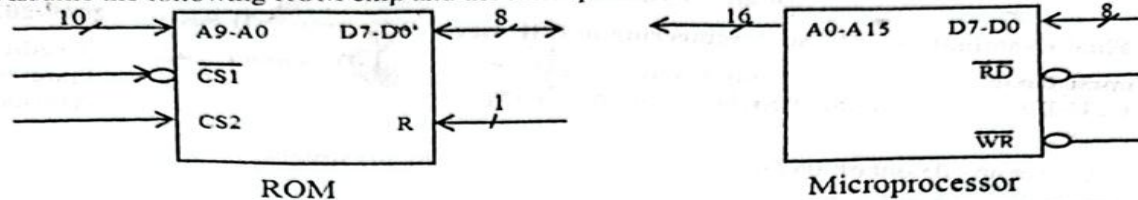
2019-2020

4[C] Convert -1313.3125_{10} to IEEE32-bit floating point format. (3)

4[D] Mention the main functions of an interface. (2)

5[A] (5)

Assume the following ROM chip and the microprocessor:



Connect the microprocessor to the ROM to obtain the memory map: 0000_{16} through $03FF_{16}$. Use only the signals shown. Draw a neat logic diagram and analyze the memory map.

2018-2019

4[C] Interpret EFE_{16} as unsigned, signed and floating-point numbers that a microprocessor can understand. (4)

5[C] (3)

The microcomputer communicates with an external device via one or more registers called I/O ports using programmed I/O. Illustrate the functions of two of them with brief description.

5[D] Make a comparison between buffers and spoolers. (2)

6[A] A processor having 16 bits address line and it is byte addressable. Interface 4096x8 RAM with the processor. (5)

- a) What is the memory capacity of the processor?
- b) How many addresses are required for a RAM?
- c) How many RAM chips are required?
- d) What will be the size of address decoder?

2017-2018

4[B] Convert 39887.5625 to IEEE 32-bit floating point format. (4)

5[B] Differentiate between programmed I/O and interrupt initiated I/O. (4)

5[C] How can the signed binary numbers be represented. (3)

5[D] What is the range of signed values that may be stored in 20-bits? (1)

6[A] How to connect a microprocessor to ROM/RAM Chips? Illustrate and explain each technique. (9)

6[B] What is an overflow flag set or reset. Give examples. (3)

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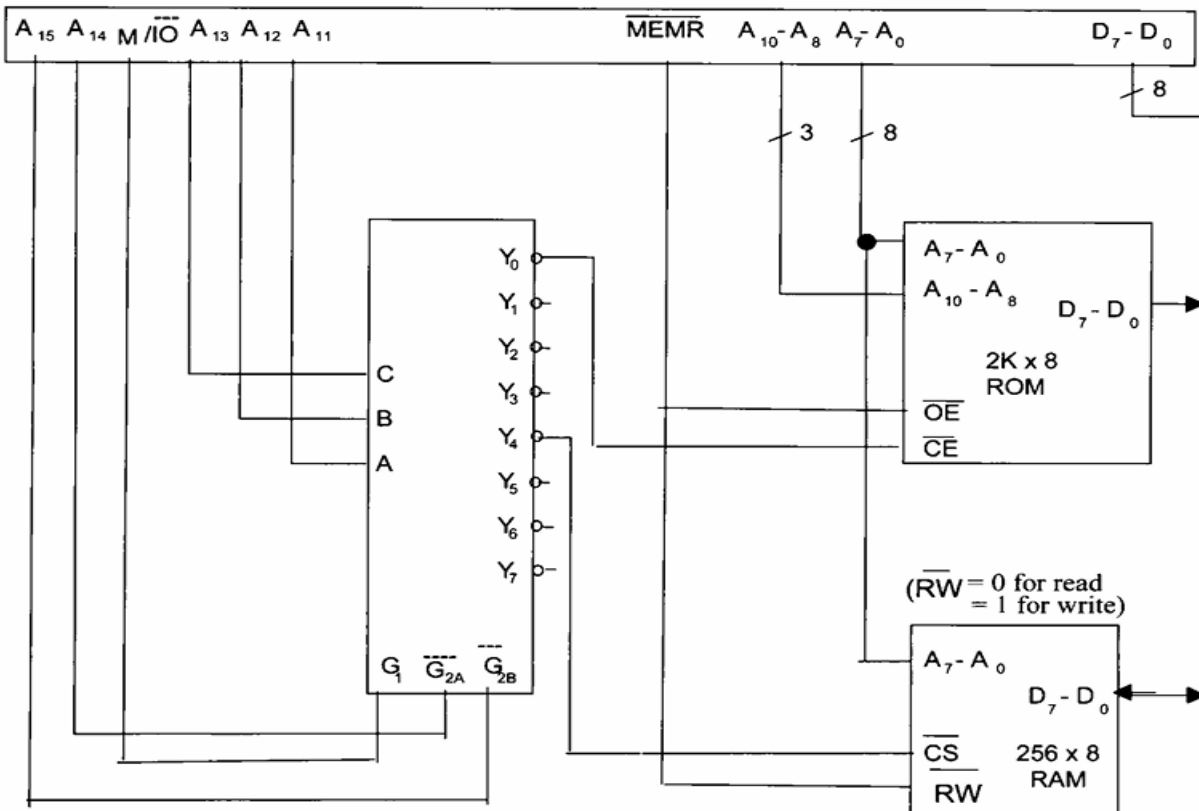
2020-2021

4[B] Given a memory with a 14-bit address and an 8-bit word size. (3)

- i. How many bytes can be stored in this memory?
- ii. If this memory were constructed from 1K x1 RAMS, how many memory chips would be required?
- iii. How many bits would be used for chip select?

5[B] Consider the hardware schematic shown in Figure. (4)

- (a) Determine the address map of this system. *Note:* $\overline{\text{MEMR}} = 0$ for read, $\overline{\text{MEMR}} = 1$ for write, $\text{M}/\overline{\text{IO}} = 0$ for I/O and $\text{M}/\overline{\text{IO}} = 1$ for memory.
- (b) Is there any possibility of bus conflict in this organization? Clearly justify your answer.



6[A] (4)

The block diagram of a 512×8 RAM chip is shown in Figure P3.11. In this arrangement the memory chip is enabled only when $\overline{CS1} = L$ and $CS2 = H$. Design a $1K \times 8$ RAM system using this chip as the building block. Draw a neat logic diagram of your implementation. Assume that the microprocessor can directly address 64K with a $R/\overline{W}$ and eight data pins. Using linear decoding and don't-care conditions as 1's, determine the memory map in hexadecimal.

